*15.095 Machine Learning Under a Modern Optimization Lens*

# Multimodal Alpha Forecasting and Robust Portfolio Optimization

**Catherine Ma** (cama12@mit.edu) & **Hanna Zhang** (zeyuzh@mit.edu)

# 1 Introduction

Short-horizon equity return prediction is one of the most challenging problems in quantitative finance. Daily returns have extremely low signal-to-noise ratios, are influenced by market microstructure noise, and arise from nonlinear macro, sector, and firm-level drivers. Despite this difficulty, even marginal improvements in predictive accuracy can lead to economically meaningful gains when integrated into a risk-controlled portfolio optimization framework.

The goal of this project is to develop a multimodal forecasting system that integrates diverse sources of information: price-based technical indicators, firm fundamentals, macroeconomic variables, sector classifications, calendar effects, and sentiment signals and text embeddings from news data. We aim to evaluate which feature groups contribute meaningful predictive power and generate a next-day return forecast suitable for prescriptive portfolio optimization.

# 2 Data

## 2.1 Data Sources

To support our modeling framework, we compile a daily panel for S&P 500 stocks using information from multiple independent datasets:

- **S&P 500 Constituents:** List of S&P 500 tickers used to define the modeling universe.
- **Stock Market Data:** Daily OHLCV (Open, High, Low, Close, Volume) prices for all S&P 500 stocks from 2016–2023 from Yahoo Finance.
- **Company Fundamentals:** Firm-level financial metrics, such as profitability, margins, growth, leverage, and valuation ratios, sourced from WRDS/Compustat.
- **Macroeconomic Indicators:** FRED variables, including interest rates, GDP growth, unemployment rates, and treasury yields.
- **News Data:** Daily financial news headlines and summaries collected from the FNSPID datasetm used to compute sentiment scores and text embeddings.[1]

## 2.2 Data Processing Pipeline

All data are merged at the (ticker, date) level to form a consistent daily panel. The main steps are:

1. **Cleaning and Alignment:** All data sources are standardized to daily frequency. To avoid look-ahead bias, missing fundamental values are forward-filled within each firm.

2. **Target Variable:** The next-day return is computed from daily closing prices and used as the prediction target.

3. **Feature Engineering:** We generate four major classes of predictors:

    - **Price-Based Technical Features:** Multi-horizon moving averages, exponential moving averages, rolling-window returns, log volume, and other short-term indicators.
    - **Fundamental Features:** Profitability metrics (ROA, ROE), margins (gross, operating, and net margins), growth rates, valuation ratios (book-to-market, earnings yield, cash-flow yield), and balance-sheet indicators (leverage, liquidity, cash holdings).

[1] https://huggingface.co/datasets/Zihan1004/FNSPID

- **Macroeconomic Variables:** VIX, yield curve spreads, Fed Funds Rate, industrial production, unemployment, GDP, and other broad economic indicators.
- **Text Features:** Sentiment scores derived from FinBERT (positive, negative, neutral classifications aggregated into daily measures such as mean sentiment, extremal sentiment, and news count), and 768-dimensional FinBERT text embeddings obtained from headlines and summaries and averaged within each stock-day to capture linguistic tone and narrative content.
- **Dimension Reduction:** High-dimensional text embeddings are compressed into 64 components via PCA, preserving most of the variance while reducing computational cost.
- **Sector and Time Structure:** One-hot sector and subsector classifications, plus day-of-week and month-of-year dummies capturing calendar seasonality.

4. **Train–Validation–Test Split:** The data is segmented chronologically into:

$$\text{Train: 2016–2021}, \quad \text{Validation: 2022}, \quad \text{Test: 2023}.$$

This prevents information leakage and ensures realistic out-of-sample evaluation.

The resulting dataset provides a multimodal view of cross-sectional and time-series drivers of equity behavior, providing a basis for evaluating the predictive contribution of different feature families.

# 3 Methodology

## 3.1 Stock Next Day Return Prediction

### Evaluation Metric

Because the goal is to forecast next-day equity returns in a noisy, low-signal environment, the evaluation framework must reflect practical trading considerations. Although models output real-valued return forecasts, the primary operational question is whether the model correctly anticipates the *direction* of the next day's movement, as this determines the trading position taken.

To align with this objective, we use **Directional Accuracy (DA)** as the evaluation metric. DA measures the fraction of predictions whose sign matches the sign of the realized next-day return:

$$\mathrm{DA} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left\{\, \mathrm{sign}(\hat{r}_i) = \mathrm{sign}(r_i) \,\right\},$$

where $\hat{r}_i$ is the predicted next-day return and $r_i$ is the realized next-day return. A DA of 50% corresponds to random guessing in a balanced up/down environment, so even small improvements (1–2 percentage points) are economically meaningful.

### 3.1.1 Baseline Models

Before introducing flexible machine learning methods, we evaluate a set of simple, deterministic forecasting rules. These baselines serve two purposes: (i) to provide interpretable benchmarks grounded in standard quantitative finance heuristics such as momentum and seasonality, and (ii) to assess how much apparent predictability can arise from rule selection rather than genuine signal. All baselines are evaluated on the same S&P 500 daily return panel using the 2016–2022 train–validation period and the 2023 test set.

**Single-Rule Baselines**

These models apply the same fixed rule across all stocks and dates:

- **Naive persistence:** predicts the next-day return using the current day's return.
- **Moving averages (MA):** simple averages over windows ranging from 3 to 60 trading days.
- **Exponential moving averages (EMA):** exponentially weighted versions of MA rules that emphasize more recent observations.
- **Seasonal rules (S1–S7):** calendar-based predictors, including same-day-last-year, same-month-last-year, month-of-year means, day-of-week means, and rolling long-horizon averages.

Across these baselines, directional accuracy remains close to the 50% benchmark, with the strongest performance coming from calendar-based rules, particularly month-of-year seasonality. This suggests that coarse calendar structure generalizes more reliably than short-horizon momentum signals.

**Hybrid Baselines**

To relax the "one-rule-for-all" constraint, we evaluate hybrid baselines that allow different segments of the data to select their preferred single-rule predictor based on 2016–2022 performance. Specifically, we consider selection at the stock level (per-ticker), at the calendar-month level (per-month), and at the weekday level (per-day-of-week).

| **Hybrid Method** | **Train+Val DA** | **Test DA** |
|---|---|---|
| Per-ticker | 0.5185 | 0.5044 |
| Per-month | 0.5144 | 0.5110 |
| Per-day-of-week | 0.5119 | 0.5040 |

Table 1: Performance of Hybrid Baseline Models

The per-ticker method achieves the highest in-sample accuracy but does not generalize well, whereas the month-of-year hybrid yields the most stable out-of-sample performance. Based on the frequency with which each rule is selected, the results suggest that calendar structure and slower-moving return components are the most consistently useful simple predictors, and these features were utilized in the subsequent machine learning models.

### 3.1.2 From Baseline Signals to Multimodal Machine Learning Models

We begin by mapping the price patterns identified by the baseline models into model features and evaluating how much predictive structure can be extracted from these signals alone. Linear regression and gradient-boosted tree models are trained on this price-only feature set, followed by feature ablation to assess the contribution of individual signals. These experiments show that a small number of price-based features captures most of the available predictive information, while larger feature sets mainly improve in-sample fit rather than out-of-sample performance.

Then, we shift focus to a set of core structural features, including calendar indicators, sector and subsector classifications, and volume-based measures. Models trained on these features deliver more stable and consistent performance than price-only specifications, highlighting the importance of calendar effects and sector-level structure at short horizons. Combining price signals with structural features provides additional flexibility, but subsequent ablation reveals that most predictive power originates from a compact subset of structural variables. In particular, day-of-week, month-of-year, and sector indicators consistently emerge as the most robust and generalizable predictors.

Finally, because ticker identifiers are excluded from the tabular feature space to avoid high-dimensional sparsity, we explore alternative approaches to incorporate firm-specific information, including neural networks with ticker embeddings and tree-based models that natively handle categorical identifiers. These experiments motivate the use of the Day–Month–Sector (DMS) feature set as the foundation for subsequent multimodal extensions.

Table 2: Directional Accuracy Across Feature Sets and Model Classes

| **Feature Set** | **Model Class** | **Val DA** | **Test DA** |
|---|---|---|---|
| Price-based signals | Linear Regression | 0.4867 | 0.4957 |
| Price-based signals | XGBoost (Top-3 Price Features) | 0.4934 | 0.5177 |
| Price-based signals | XGBoost (Top-15 Price Features) | 0.5016 | 0.5168 |
| Core structural features | Linear Regression | 0.4805 | 0.5132 |
| Core structural features | XGBoost | 0.5166 | 0.5214 |
| Price signals + structural features | XGBoost | 0.5136 | 0.5139 |
| DMS | XGBoost | 0.5146 | 0.5281 |
| DMS | LightGBM | 0.5169 | 0.5280 |
| DMS + Ticker embeddings | Neural Network | 0.4931 | 0.5199 |
| DMS + Categorical Ticker | CatBoost | 0.5179 | 0.5354 |

#### 3.1.3 Extending DMS with Additional Feature Families

Having identified day-of-week, month-of-year, and sector indicators (DMS) as the most robust and generalizable core predictors, we next evaluate whether incorporating additional sources of information can further improve predictive performance. We extend the DMS feature set by incrementally adding firm fundamentals, macroeconomic variables, and news-based sentiment features. All feature extensions are evaluated using tree-based gradient boosting methods (XGBoost, CatBoost, and LightGBM), which consistently outperform linear and neural models in earlier stages of the analysis.

**Results**

Extending the DMS core with additional feature families yields heterogeneous effects. Adding firm-level fundamentals does not improve performance and often reduces both validation and test directional accuracy, suggesting that slow-moving accounting variables add little incremental information at a one-day horizon and may introduce noise. Macroeconomic variables similarly offer limited gains once calendar and sector structure are controlled for, as their effects appear largely absorbed by the DMS features.

In contrast, news-based sentiment features provide meaningful incremental value when summarized using simple aggregates such as mean sentiment. More complex sentiment dynamics and high-dimensional text embeddings do not yield further improvements and can degrade performance, indicating unfavorable signal-to-noise trade-offs at short horizons.

Among all configurations evaluated, LightGBM consistently delivers the strongest performance. Two specifications are particularly notable. The model achieving the best validation performance combines the DMS feature set with a small subset of firm-level fundamental variables, capturing valuation and profitability characteristics. In contrast, the model achieving the strongest out-of-sample performance augments DMS with mean news sentiment, indicating that aggregate shifts in news tone provide incremental predictive value beyond structural features alone.

Table 3: Best-Performing Models

| **Feature Set** | **Model Class** | **Val DA** | **Test DA** |
|---|---|---|---|
| DMS + Selected Fundamentals | LightGBM | **0.5218** | 0.5272 |
| DMS + Mean Sentiment | LightGBM | 0.5143 | **0.5398** |

## 3.2 Robust Portfolio Optimization

While predictive accuracy indicates that the model captures return-relevant patterns, turning point forecasts into actionable trades requires an allocation rule that respects risk, diversification, and transaction costs. Simple heuristics such as equal-weighting or top-$K$ selection ignore downside exposures and often produce unstable, overly concentrated portfolios whose performance exaggerates the value of the underlying signal.

To obtain realistic and risk-aware allocations, we employ a robust Conditional Value-at-Risk (CVaR) optimization layer. CVaR determines the weight assigned to each equity by evaluating how the portfolio would behave under a rolling set of recent historical return scenarios. For each day's prediction vector, the optimizer searches for weights that increase exposure to stocks with high predicted returns, but penalize those that contribute heavily to losses in the worst tail of the return distribution. In practice, this yields portfolios that favor assets with both attractive forecasts and stable downside behavior, while enforcing constraints such as long-only positions, position-size caps, and turnover limits. The result is a set of portfolio weights that are smoother, more diversified, and more robust than those produced by naïve allocation rules.

### 3.2.1 CVaR Formulation

Rather than relying on variance as a proxy for risk, we use CVaR to focus directly on large downside losses. CVaR measures the expected loss in the worst $(1-\alpha)$ fraction of outcomes and remains well-behaved even when returns are skewed or heavy-tailed. Given portfolio weights $w$ and a set of recent scenario returns $\{r_t\}_{t=1}^{T}$, CVaR at confidence level $\alpha = 0.95$ is written using auxiliary variables $v$ (the VaR cutoff) and $z_t$ (tail losses):

$$\mathrm{CVaR}_\alpha(w) = v + \frac{1}{(1-\alpha)T}\sum_{t=1}^{T} z_t, \qquad z_t \geq 0, \quad z_t \geq -(r_t^\top w) - v.$$

This linear representation makes the CVaR objective directly optimizable.

### 3.2.2 Rolling Scenario Construction

The scenario matrix is built from the most recent 60 days of realized cross-sectional returns. Using a rolling window allows the optimizer to reflect current market conditions—including changes in volatility, correlations, and downside risk—without assuming a particular return distribution. As market regimes shift, the CVaR optimizer naturally updates its estimate of tail risk and adjusts portfolio weights accordingly.

### 3.2.3 Optimization Objective and Constraints

At each date $t$, the optimizer selects weights for each equity by solving

$$\max_{w} \quad \hat{\mu}_t^\top w \;-\; \lambda\,\mathrm{CVaR}_{0.95}(w) \;-\; \mathrm{TC}\cdot\|w-w_{t-1}\|_1,$$

where $\hat{\mu}_t$ are the model's predicted next-day returns (scaled for numerical stability), $\lambda$ controls aversion to tail risk, and the $\ell_1$ turnover term penalizes excessive rebalancing and incorporates proportional transaction costs.

We impose the following practical trading constraints:

- **Long-only and fully invested:** $w_i \geq 0$, $\sum_i w_i = 1$.
- **Position limit:** $w_i \leq 2\%$ to prevent concentration.
- **Turnover limit:** $\|w - w_{t-1}\|_1 \leq 10\%$ to avoid hyperactive trading.
- **Transaction cost penalty:** proportional cost of 5 bps per unit turnover.

These constraints make the optimization problem both realistic and stable, preventing extreme weight allocations that often arise in unconstrained ML portfolios.

### 3.2.4 Backtesting Procedure

For each model's cross-sectional predictions, we build a $T \times N$ matrix of predicted returns and a matching matrix of realized one-day-ahead returns. The portfolio is rebalanced daily using the CVaR optimizer. Performance is reported using standard metrics:

Sharpe ratio, annualized return, annualized volatility, and maximum drawdown.

Importantly, CVaR-based optimization acts as a "stress test" for the machine learning signals: only predictions that are stable, consistent across the cross-section, and resistant to noise will survive the downside-risk penalty and turnover constraints.

# 4 Results

To compare the predictive contribution of different input modalities, we evaluate five LightGBM models trained on progressively richer feature sets:

- **Fundamentals:** Day-Month-Sector (DMS), log market capitalization ($\log\_mktcap$), book-to-market ratio (bm), return on assets (roa_ttm), and return on equity (roe_ttm).
- **Mean Sentiment:** DMS, and mean sentiment score from news data.
- **Mean Sentiment + Embeddings:** DMS, mean sentiment score, and text embeddings.
- **News Momentum:** DMS and a full set of leak-free sentiment momentum predictors. For each raw sentiment variable $f \in \{\mathrm{mean}, \mathrm{max}, \mathrm{min}, \mathrm{sum}, \mathrm{count}\}$, the feature set includes: $f$, $f_{\mathrm{lag1}}$, $f_{\mathrm{roll3}}$, $f_{\mathrm{roll5}}$, $f_{\mathrm{vol5}}$, $f_{\mathrm{shock}}$, and $f_{\mathrm{surprise3}}$.
- **News Momentum + Embeddings:** DMS, News Momentum, and text embeddings.

## Directional Accuracy for LightGBM Models with Different Feature Sets

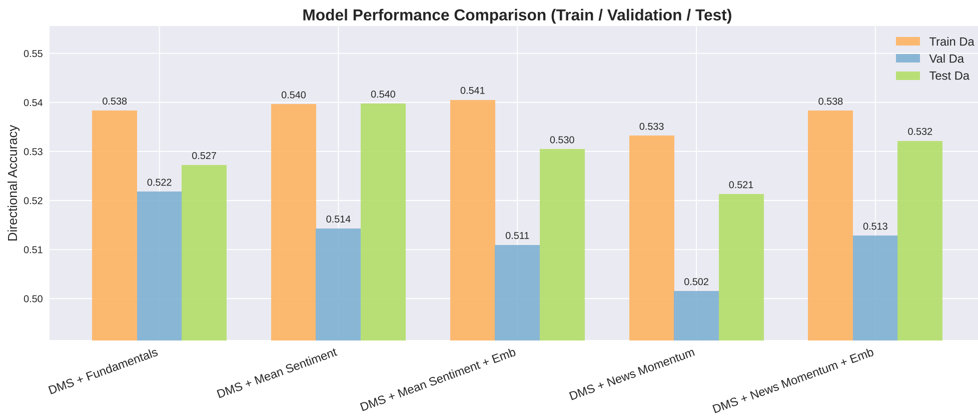


Figure 1: Directional accuracy for LightGBM models trained with different feature sets.

**Interpretation.** Across all models, directional accuracy remains in the narrow range of 0.51–0.54, consistent with well-known limits on daily return predictability. All specifications outperform a random-guessing benchmark, indicating that even simple tabular signals contain economically meaningful information. Models incorporating fundamentals or mean news sentiment achieve the strongest test performance, suggesting that slow-moving valuation measures and aggregate sentiment are the most stable predictors. In contrast, PCA-based text embeddings do not improve accuracy and often degrade it, implying an unfavorable signal-to-noise trade-off at daily horizons. Richer news momentum features provide limited incremental gains and exhibit signs of overfitting.

## Portfolio Performance under CVaR Optimization with Different Feature Sets

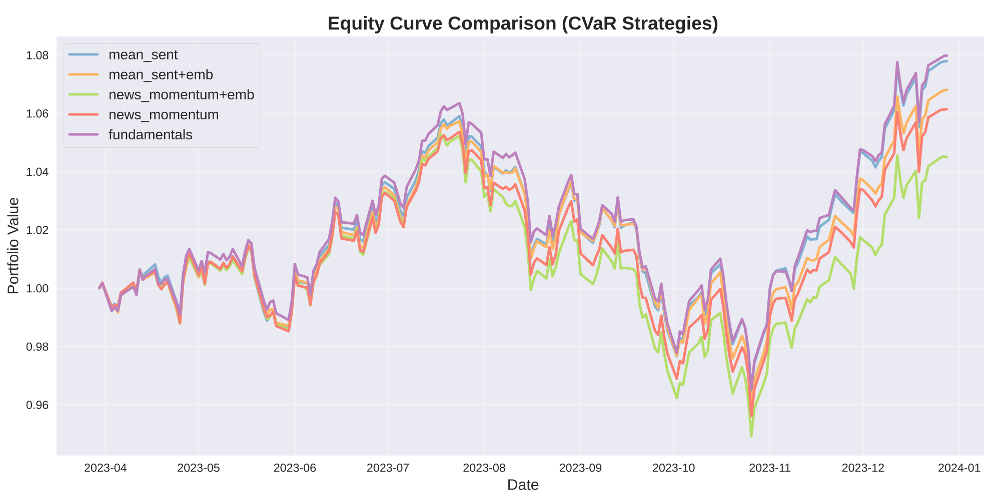


Figure 2: Equity curves for portfolios using CVaR optimization with different feature sets.

| **Model** | **Sharpe** | **Ann. Return (%)** | **Mean IC** | **IR** |
|---|---|---|---|---|
| DMS + Fundamentals | **1.166** | **10.84** | **0.02834** | **0.1669** |
| DMS + Mean Sentiment | **1.142** | **10.58** | **0.02685** | **0.1691** |
| DMS + Mean Sentiment + Embeddings | 0.989 | 9.23 | 0.02067 | 0.1629 |
| DMS + News Momentum | 0.895 | 8.32 | 0.02786 | 0.1906 |
| DMS + News Momentum + Embeddings | 0.654 | 6.07 | 0.01457 | 0.1312 |

Table 4: Performance and Information Coefficient (IC) Statistics for Different Feature Sets

**Interpretation.** Table 4 summarizes both economic performance and statistical signal quality. The *Sharpe ratio* and *annualized return* summarize portfolio performance, while the *Information Coefficient (IC)* and *Information Ratio (IR)* measure the cross-sectional predictive strength and stability.

Fundamentals and mean sentiment deliver the strongest performance, with the highest Sharpe ratios ($\approx$ 1.14–1.17) and mean IC values ($\approx$ 0.027–0.028). This suggests that slow-moving fundamentals and aggregate sentiment remain the most reliable predictors at a daily horizon. Adding text embeddings consistently reduces both Sharpe and IC, indicating that high-dimensional embeddings introduce noise rather than improving predictability. News momentum features offer moderate improvements in IC stability (higher IR) but do not outperform fundamentals in realized portfolio performance. Overall, simpler feature sets provide clearer and more stable signals than embedding-augmented models.

## 5 Discussion and Conclusion

This project studies next-day equity return prediction in a setting where predictability is known to be weak and noise dominates at daily horizons. Our results reinforce this: across baseline and flexible machine learning models, directional accuracy remains close to the 50% benchmark, and gains are incremental rather than dramatic. Nevertheless, the stability of these improvements across model classes and feature sets suggests that the extracted signals are not purely artifacts of model selection and can yield economic value when paired with a risk-aware allocation framework.

A central finding is the importance of structural features. Day-of-week, month-of-year, and sector indicators consistently emerge as the most robust predictors, while many price-based signals that improve in-sample fit fail to generalize once structural variables are included. This reflects the fragility of short-horizon price patterns and the greater stability of calendar and sector effects.

Adding further modalities produces mixed results. Macroeconomic variables provide little incremental benefit beyond structural features, whereas simple news-based sentiment measures, such as mean sentiment, offer meaningful out-of-sample improvements. More complex sentiment dynamics and high-dimensional text embeddings do not improve performance and often degrade it, indicating that additional complexity introduces noise rather than usable signal at a one-day horizon.

While predictive accuracy provides a useful diagnostic, our results show that economic value depends on how forecasts are converted into trades. The robust CVaR optimization layer plays a critical role by penalizing tail risk and turnover, producing smoother and more diversified portfolios than naïve allocation rules. In backtests, models built on compact, stable feature sets (DMS with fundamentals or mean sentiment) deliver the strongest risk-adjusted performance, with the highest Sharpe ratios and the most reliable information coefficients. Embedding-augmented models underperform both statistically and economically, further supporting the conclusion that simpler representations yield clearer signals in this setting.

Taken together, the results suggest a practical modeling principle for short-horizon equity forecasting: prioritize compact, high-bias feature sets that capture persistent structure, and add new modalities only when they provide stable incremental information. Among the models considered, tree-based boosting methods, particularly LightGBM, provide the best combination of performance and stability on tabular financial data. Overall, although next-day return prediction remains extremely challenging, the combination of modestly informative multimodal signals with robust CVaR-based portfolio construction yields economically meaningful improvements in risk-adjusted performance.

# 6 Contribution

- **Catherine:** Prediction Models
- **Hanna:** Data Preparation and Portfolio Optimization